GrIPS: Gradient-free, Edit-based Instruction Search for Prompting Large Language Models

Archiki Prasad, Peter Hase, Xiang Zhou, Mohit Bansal · UNC Chapel Hill · arXiv 2203.07281v2

HarnessRSI 67 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

Only the instruction text is edited (weights frozen)

- **del** delete a phrase
- **swap** swap two phrases
- **par** paraphrase a phrase
- **add** add back a phrase

1. **Split into phrases** — parser splits the original instruction
2. **5 candidates** — each gets 1 random edit (above)
3. **Score on 100 non-test examples** — balanced acc. + 10 × entropy
4. **Keep rule** — best > base score → new base, else keep
5. **Stop** — patience P = 2 or 10 iterations

≤10×

final instruction → evaluated on held-out test
kept artifact = the searched instruction text

2 Headline: +9.36 / +4.29 / +2.36 points

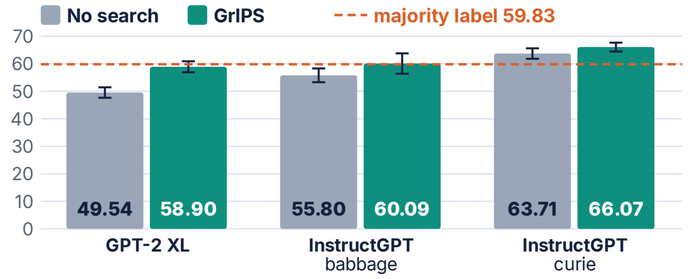


Table 1: 8 classification tasks, 300 test examples per task, 3 search seeds; whiskers = 95% CI.

3 Controls that decide if the gain is real

Ablation · GPT-2 XL (Table 2, single numbers)



Drops 1.01–2.56; Table 1 CIs 1.6–3.7; Tables 2–6 have no variance.

Table 3 · 100 test/task	GPT-2 XL	babbage	curie
No search	48.38	55.37	57.25
Manual rewrite	**47.70**	55.50	57.87
Manual + label line	48.12	55.37	55.37
Example search, same queries	**56.00**	56.25	57.75
GrIPS	**53.68**	**57.79**	**59.37**

Example search (Examples-Only prompts): same score set and max queries. Rewrites by the authors, ~5 h.

GPT-2 XL (Table 4)	params tuned	acc.
Direct finetuning	100%	55.88
Adapters	3%	55.08
Prefix-tuning	3%	53.29
GrIPS greedy	0	**53.68**
GrIPS beam B=5 (≈5× evals)	0	**56.50**

Gradient methods: test input only (fn 11), 80:20 split of score set; tuning and compute not reported.

4 Where the gain is absent or thin

- **GPT-2 XL ends at 58.90**, below the majority-label 59.83; babbage 60.09 (CI 3.7); only curie (66.07) is clearly above
- **Concentrated in few tasks** (read off Fig. 4): GPT-2 XL's +9.36 is mostly tasks 022, 195, 021; about half of babbage's +4.29 is task 021; curie is lower after search on 021 and 022
- Tasks improved (of 8): GPT-2 XL **4**, babbage 7, curie 5 (Table 10); plain manual rewriting on GPT-2 XL is below no search
- Example search beats GrIPS on GPT-2 XL (56.00 vs 53.68); finetuning and adapters beat greedy GrIPS
- Only 8 binary tasks; hyperparameters tuned on unnamed tasks. Text: 4.27 gain at $|S| = 100$; Table 3: 2.42‡

Takeaway

Random phrase edits kept by a 100-example score set raise average accuracy 2.36–9.36 points, mostly on a few tasks. On InstructGPT models the averages are 1.50–2.29 above manual rewriting and same-query example search (no CIs); on GPT-2 XL example search is higher.

Our read · HarnessRSI

Score 67 · ★★ · evidence level: retained

Reports: held-out test, 3 seeds, 95% CIs in Table 1 only, same-query example search, ablations, cost.

Does not report: leakage check, transfer, CIs in Tables 2–6.